CS 3244
Machine Learning

Concept Learning

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2 post

Chapter 2, Machine Learning (Mitchell)

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Recap and Forecast

Learning is used when

1. A pattern exists
2. We cannot pin it down mathematically
3. We have data on it

Focus on supervised learning

- Unknown target function
  $$y = f(\mathbf{x})$$
- Data set
  $(\mathbf{x}^{(1)}, y^{(1)}), \ldots, (\mathbf{x}^{(m)}, y^{(m)})$
- Learning algorithm picks $h_\theta \approx f$ from hypothesis set $\mathcal{H}$

Learning Outcomes for this week:

- Learning from Examples
- General-to-Specific ordering over hypotheses spaces
- Version spaces
- Candidate Elimination Algorithm
- Picking new examples
- The need for inductive bias

Version Spaces

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Version Spaces

Definition. The version space $VS_{\mathcal{H},\mathbf{D}}$ w.r.t. hypothesis space $\mathcal{H}$ and training examples $\mathbf{D}$, is the subset of hypotheses from $\mathcal{H}$ that are consistent with $\mathbf{D}$:

Remember, $\mathcal{H}$ denotes the set of all hypotheses for a particular learner A.

$$VS_{\mathcal{H},\mathbf{D}} = \{h \in \mathcal{H} \mid h \text{ is consistent with } \mathbf{D}\}.$$

- If $f \in \mathcal{H}$, then a large enough $\mathbf{D}$ can reduce $VS_{\mathcal{H},\mathbf{D}}$ to $\{f\}$.
- If $\mathbf{D}$ is insufficient, then $VS_{\mathcal{H},\mathbf{D}}$ represents the uncertainty of what the target concept is.
- $VS_{\mathcal{H},\mathbf{D}}$ contains all consistent hypotheses, including maximally specific hypotheses.

List-Then-Eliminate Algorithm

"Make a list and check it twice"

Intuition. List all hypotheses in $\mathcal{H}$. Then, eliminate any hypothesis found inconsistent with any training example.

1. *VersionSpace* $\leftarrow$ a list containing every hypothesis in $\mathcal{H}$.

2. For each training example $\langle \mathbf{x}, f(\mathbf{x}) \rangle$.

   Remove from *VersionSpace* any hypothesis h for which $h(\mathbf{x}) \neq f(\mathbf{x})$.

3. Output the list of hypotheses in *VersionSpace*.

Limitation. Totally unrealistic, requires full search.
Prohibitively expensive to exhaustively enumerate all hypotheses for most finite $\mathcal{H}$.

Version Space for *EnjoyDrink*

State after observing the 4 training examples from Pre, Slide 9

S: {<Kopi, Si, ?, Peng, ?, ?> }

<Kopi,?, ?, Peng, ?, ?> <Kopi, Si, ?, ?, ?, ?> <?, Si, ?, Peng, ?, ?>

G: {<Kopi, ?, ?, ?, ?, ?>, <?, Si, ?, ?, ?, ?> }

Compact Representation of Version Space

Definition. The general boundary G of $VS_{\mathcal{H},\mathbf{D}}$ is the set of maximally general members of $\mathcal{H}$ consistent with $\mathbf{D}$:

$$G = \{g \in \mathcal{H} \mid g \text{ consistent with } \mathbf{D} \wedge (\neg\exists g' \in \mathcal{H} \quad g' >_g g \wedge g' \text{ consistent with } \mathbf{D})\}.$$

Definition. The specific boundary S of $VS_{\mathcal{H},\mathbf{D}}$ is the set of maximally specific members of $\mathcal{H}$ consistent with $\mathbf{D}$:

$$S = \{s \in \mathcal{H} \mid s \text{ consistent with } \mathbf{D} \wedge (\neg\exists s' \in \mathcal{H} \quad s >_g s' \wedge s' \text{ consistent with } \mathbf{D})\}.$$

Every member of the version space lies between these boundaries.

Version space representation theorem

$$VS_{\mathcal{H},\mathbf{D}} = \{h \in \mathcal{H} \mid \exists s \in S \;\; \exists g \in G \quad g \geq_g h \geq_g s\}.$$

Proof of the Version Space Representation Theorem

$\Leftarrow$ Every h satisfying RHS is in $VS_{\mathcal{H},\mathbf{D}}$.

1. Choose arbitrary $g \in G, s \in S, h \in \mathcal{H}$ such that $g \geq_g h \geq_g s$
2. Every +ve training instance satisfies s, by Def. of S and Prop. 1
3. Since $h \geq_g s$, every +ve training instance satisfies h
4. Every −ve training instance does not satisfy g, by Def. of G and Prop. 1
5. Since $g \geq_g h$, every −ve training instance does not satisfy h
6. h is consistent with $\mathbf{D}$, by Prop. 1
7. $h \in VS_{\mathcal{H},\mathbf{D}}$

Your Turn: $\Rightarrow$ Every member of $VS_{\mathcal{H},\mathbf{D}}$ satisfies RHS.

Hint: assume some h in $VS_{\mathcal{H},\mathbf{D}}$ doesn't satisfy the RHS expression, then show that leads to inconsistency.

Candidate Elimination

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Candidate-Elimination Algorithm

Intuition. Start with most general and specific hypotheses. Each training example "minimally" generalizes S and specializes G to remove inconsistent hypotheses from version space.

1. $G \leftarrow$ maximally general hypotheses in H

2. $S \leftarrow$ maximally specific hypotheses in H

Candidate-Elimination Algorithm

3. For each training example $\mathbf{x}$:

If $\mathbf{x}$ is a +ve example, /* then generalize S if necessary */

- Remove from G any hypothesis inconsistent with $\mathbf{x}$.
- For each $s \in S$ not consistent with $\mathbf{x}$:
  - Remove s from S.
  - Add to S all minimal <u>generalizations</u> h of s such that
    - h is consistent with $\mathbf{x}$, and
    - some member of G is more <u>general</u> than h.
  - Remove from S any hypothesis that is more <u>general</u> than another hypothesis in S

Candidate-Elimination Algorithm

/* Influence of +ve and −ve examples on S & G are dual */

If $\mathbf{x}$ is a −ve example, /* then specialize G if necessary */

- Remove from S any hypothesis inconsistent with $\mathbf{x}$.
- For each $g \in G$ not consistent with $\mathbf{x}$.
  - Remove g from G.
  - Add to G all minimal <u>specializations</u> h of g such that
    - h is consistent with $\mathbf{x}$, and
    - some member of S is more <u>specific</u> than h.
  - Remove from G any hypothesis that is more <u>specific</u> than another hypothesis in G.

Candidate-Elimination Trace (1/3)

+ve examples

Remove from G any hypothesis inconsistent with $\mathbf{x}$.

- For each $s \in S$ not consistent with $\mathbf{x}$:
  - Remove s from S.
  - Add to S all minimal generalizations h of s such that
    - h is consistent with $\mathbf{x}$, and
    - some member of G is more general than h.
  - Remove from S any hypothesis that is more general than another hypothesis in S.

$S_0 =$ <∅, ∅, ∅, ∅, ∅, ∅>

$S_1 =$ <Kopi, Si, Kosong, Peng, Gau, Lim>

$S_2 =$ <Kopi, Si, ?, Peng, Gau, Lim>

Training Examples:

$x^{(1)}$ = <Kopi, Si, Kosong, Peng, Gau, Lim>, +

$x^{(2)}$ = <Kopi, Si, Ga Dai, Peng, Gau, Lim>, +

G_0, G_1, G_2: <?, ?, ? , ?, ?, ?>

Candidate-Elimination Trace (2/3)

−ve examples

Remove from S any hypothesis inconsistent with $\mathbf{x}$.

- For each $g \in G$ not consistent with $\mathbf{x}$:
  - Remove g from G.
  - Add to G all minimal specializations h of g such that
    - h is consistent with $\mathbf{x}$, and
    - some member of S is more specific than h.
  - Remove from G any hypothesis that is more specific than another hypothesis in G

S_2, S_3 = <Kopi, Si, ?, Peng, Gau, Lim>

G_3 = <Kopi, ?, ?, ?, ?, ?> <?, Si, ?, ?, ?, ?> <?, ?, ?, ?, ?, Lim>

G_2: <?, ?, ? , ?, ?, ?>

Training Examples:

$x^{(1)}$ = <Kopi, Si, Kosong, Peng, Gau, Lim>, +

$x^{(2)}$ = <Kopi, Si, Ga Dai, Peng, Gau, Lim>, +

$x^{(3)}$ = <Teh, O, Gai Dai, Peng, Gau, Tabao>, −

Candidate-Elimination Trace (3/3)

+ve examples

Remove from G any hypothesis inconsistent with $\mathbf{x}$.

- For each $s \in S$ not consistent with $\mathbf{x}$:
  - Remove s from S.
  - Add to S all minimal generalizations h of s such that
    - h is consistent with $\mathbf{x}$, and
    - some member of G is more general than h.
  - Remove from S any hypothesis that is more general than another hypothesis in S.

S_3 = <Kopi, Si, ?, Peng, Gau, Lim>

S_4 = <Kopi, Si, ?, Peng, ?, ?>

G_4 = <Kopi, ?, ?, ?, ?, ?> <?, Si, ?, ?, ?, ?>

G_3 = <Kopi, ?, ?, ?, ?, ?> <?, Si, ?, ?, ?, ?> <?, ?, ?, ?, ?, Lim>

Training Examples:

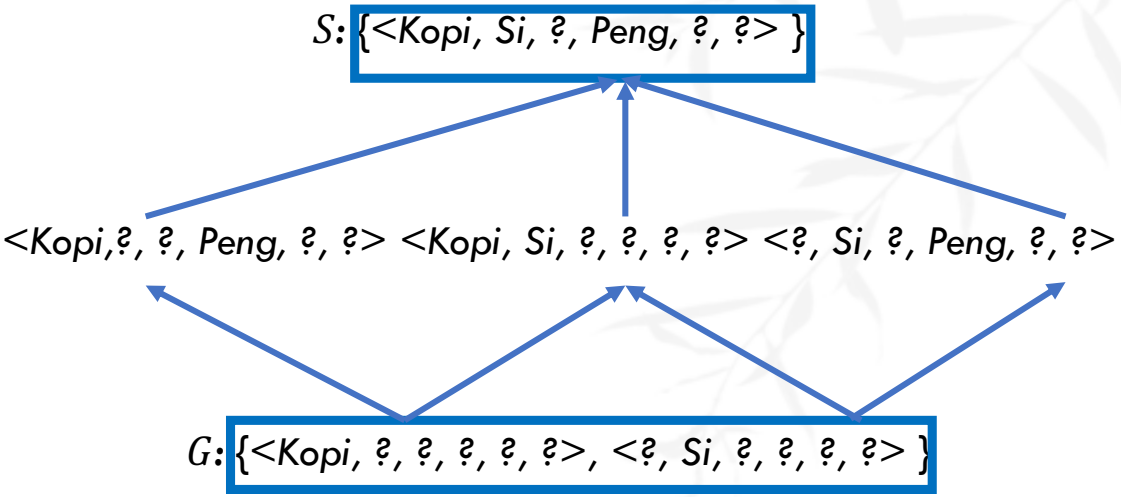
$x^{(1)}$ = <Kopi, Si, Kosong, Peng, Gau, Lim>, +

$x^{(2)}$ = <Kopi, Si, Ga Dai, Peng, Gau, Lim>, +

$x^{(3)}$ = <Teh, O, Gai Dai, Peng, Gau, Tabao>, –

$x^{(4)}$ = <Kopi, Si, Ga Dai, Peng, Por, Tabao>, +

Final Version Space for *EnjoyDrink*



Training Examples:

$x^{(1)}$ = <Kopi, Si, Kosong, Peng, Gau, Lim>, +
$x^{(2)}$ = <Kopi, Si, Ga Dai, Peng, Gau, Lim>, +
$x^{(3)}$ = <Teh, O, Gai Dai, Peng, Gau, Tabao>, –
$x^{(4)}$ = <Kopi, Si, Ga Dai, Peng, Por, Tabao>, +

Properties of Candidate Elimination

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Properties of Candidate-Elimination

Error/noise in training data

e.g., 2nd training example wrongly labeled as −ve

- Hypotheses inconsistent with 2nd example removed (including target concept f)
- S and G reduced to $\emptyset$ with sufficiently large data

Training Examples:

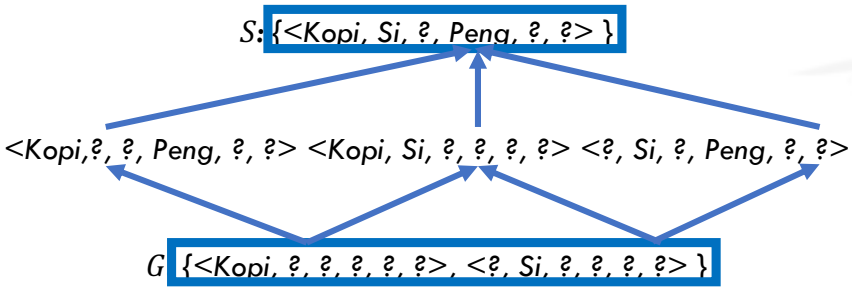
$x^{(1)}$ = <*Kopi, Si, Kosong, Peng, Gau, Lim*>, +
$x^{(2)}$ = <*Kopi, Si, Ga Dai, Peng, Gau, Lim*>, ~~+~~ −
$x^{(3)}$ = <*Teh, O, Gai Dai, Peng, Gau, Tabao*>, −
$x^{(4)}$ = <*Kopi, Si, Ga Dai, Peng, Por, Tabao*>, +

Insufficiently expressive hypothesis representation $\rightarrow f \notin H$?

S and G also reduced to $\emptyset$ with sufficiently large data

Active Learning in Candidate-Elimination

Suppose we have the hypothesis space:



What instance should an active learner request to best discriminate among the different hypotheses?

(e.g., $\mathbf{x} = \langle$*Kopi, Si, Kosong, –, Gau, Lim*$\rangle$?)

In general: A good strategy is to generate examples that satisfy exactly half of hypotheses in version space (where possible).

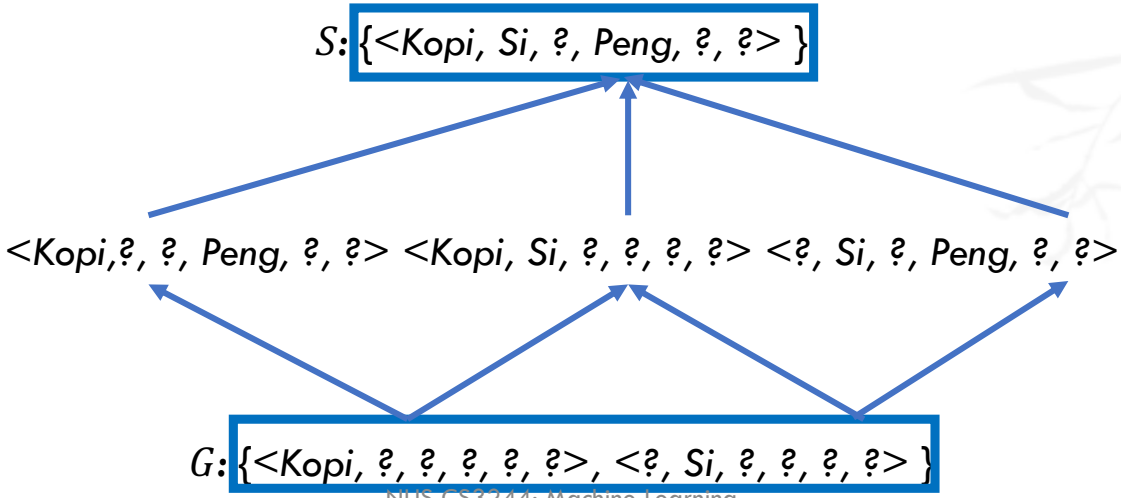
- Version space reduces by half with each training example, hence requiring at least $\lceil \log_2(VS_{\mathcal{H},\mathbf{D}}) \rceil$ examples to find target concept f.

Properties of Candidate-Elimination

How to classify new unobserved input instance? What degree of confidence?

Test instance $x^{(1)} = \langle$ *Kopi, Si, Kosong, Peng, Por, Tabao* $\rangle$

Proposition 3. An input instance $\mathbf{x}$ satisfies every hypothesis in $VS_{\mathcal{H},\mathbf{D}}$ iff $\mathbf{x}$ satisfies every member of S.



Proof of Proposition 3

$\Leftarrow$ Every input instance $\mathbf{x}$ that satisfies every $s \in S$ also satisfies every $h \in VS_{\mathcal{H},\mathbf{D}}$.

1. $\forall s \in S \quad s(x) = 1$ is given
2. $\forall h \in VS_{\mathcal{H},\mathbf{D}} \quad \exists s \in S \quad h \geq_g s$, by VSRT
3. $\forall h \in VS_{\mathcal{H},\mathbf{D}} \quad \exists s \in S \quad (s(\mathbf{x}) = 1) \rightarrow (h(\mathbf{x}) = 1)$, by Def. of $\geq_g$
4. $\forall h \in VS_{\mathcal{H},\mathbf{D}} \quad h(\mathbf{x}) = 1$, by Steps 1 and 3
5. $\therefore \mathbf{x}$ satisfies every $h \in VS_{\mathcal{H},\mathbf{D}}$

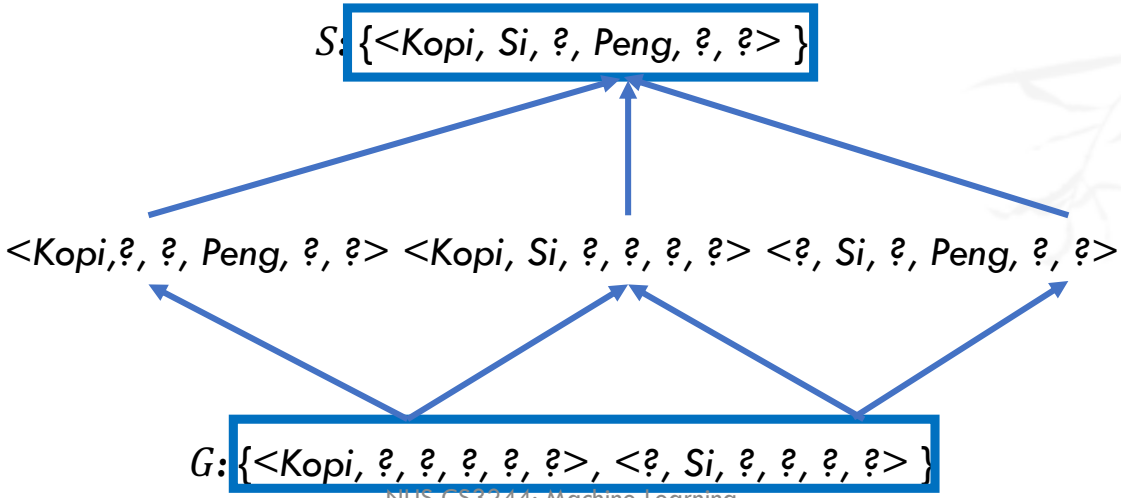
Your turn:
$\Rightarrow$ Every input instance that satisfies every $h \in VS_{\mathcal{H},\mathbf{D}}$ also satisfies every $s \in S$.

Properties of Candidate-Elimination

How to classify new unobserved input instance? What degree of confidence?

Test instance $x^{(2)} = \langle$*Teh, O, Kosong, –, Gau, Lim*$\rangle$

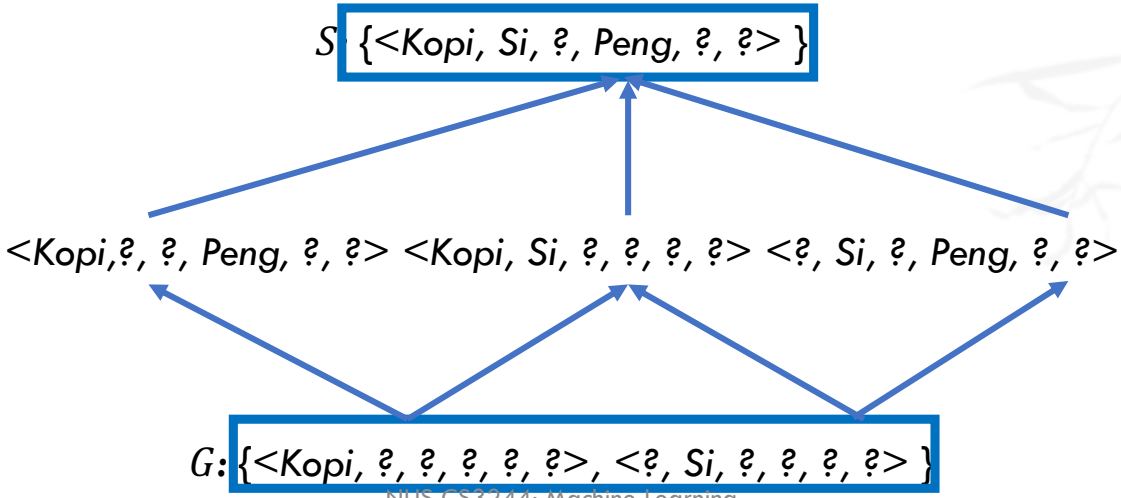
Proposition 4. An input instance $\mathbf{x}$ satisfies none of the hypotheses in $VS_{\mathcal{H},\mathbf{D}}$ iff $\mathbf{x}$ satisfies none of the members of G.



Properties of Candidate-Elimination

How to classify new unobserved input instance? What degree of confidence?

Test instance $x^{(3)} = \langle Kopi, Si, Kosong, -, Gau, Lim \rangle$

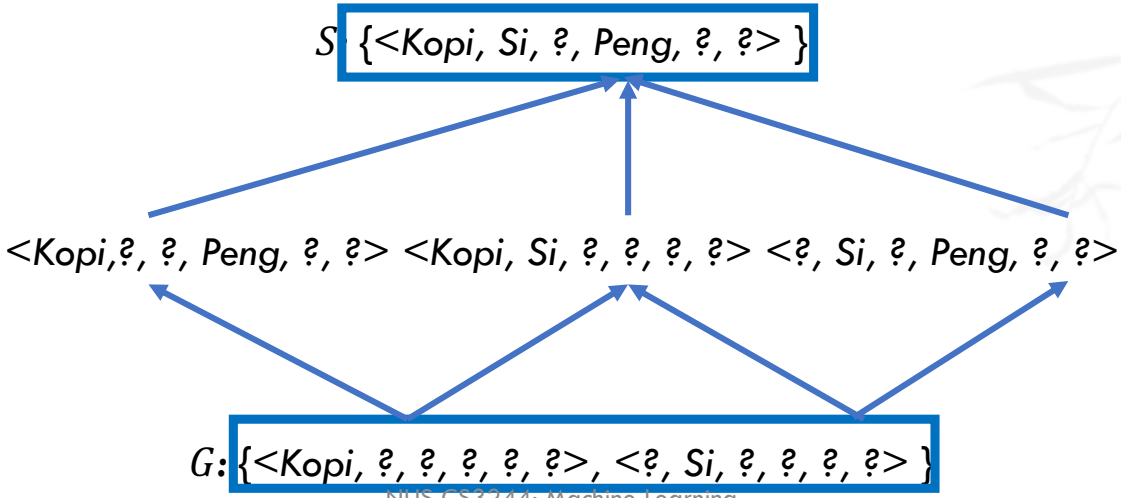


Properties of Candidate-Elimination

How to classify new unobserved input instance? What degree of confidence?

Test Instance $x^{(4)} = \langle$*Teh, Si, Kosong, Peng, Gau, Lim*$\rangle$

Majority vote is the most probable classification, assuming all hypotheses in $\mathcal{H}$ are equally probable *a priori*.



Inductive Bias

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An Unbiased Learner

Intuition. Choose $\mathcal{H}$ that can express every teachable concept (i.e., $\mathcal{H}$ is the power set of $\mathcal{X}$)

1. Consider $\mathcal{H}' \equiv$ disjunctions, conjunctions, negations of our earlier hypotheses in $\mathcal{H}$ for *EnjoyDrink* task:

Then for the example data: $\langle x^{(1)}, +\rangle, \langle x^{(2)}, +\rangle, \langle x^{(3)}, +\rangle, \langle x^{(4)}, -\rangle, \langle x^{(5)}, -\rangle$

$S \leftarrow ?$

$G \leftarrow ?$

2. Need training examples for every possible input instance in $\mathcal{X}$ to converge to the target concept

Limitation. Cannot classify new unobserved input instances (i.e., generalize beyond observed training examples)

Inductive Bias

Given

- Concept learning algorithm A
- Input space $\mathcal{X}$, unknown target concept f
- Noise-free training examples $\mathbf{D}_f = \{\langle x^{(k)}, f(x^{(k)})\rangle\}_{k=1,\ldots,m}$

(in the original text, A is written as L for learner)

Let $A(\mathbf{x}, \mathbf{D}_f)$ denote the classification of input instance $\mathbf{x}$ by A after learning from training examples $\mathbf{D}_f$.

Definition. The inductive bias of A is any minimal set of assertions B such that for any target concept f and corresponding training examples $\mathbf{D}_f$,

$$\forall \mathbf{x} \in \mathcal{X}\ (B \wedge \mathbf{D}_f \wedge \mathbf{x}) \vDash (f(\mathbf{x}) = A(\mathbf{x}, \mathbf{D}_f)) .$$

Inductive Bias of Candidate-Elimination

Inductive bias of Candidate-Elimination. $B = \{f \in \mathcal{H}\}$.

Assumption. Candidate-Elimination outputs a classification $A(\mathbf{x}, \mathbf{D}_f)$ of input instance $\mathbf{x}$ if this vote among hypotheses in $VS_{\mathcal{H},\mathbf{D}_f}$ is unanimously +ve or −ve, and does not output a classification otherwise.

1. If $f \in \mathcal{H}$, then $f \in VS_{\mathcal{H},\mathbf{D}_f}$, since f is consistent with $\mathbf{D}_f$, by Def. of version space.
2. If A outputs $A(\mathbf{x}, \mathbf{D}_f)$, then $h(\mathbf{x}) = A(\mathbf{x}, \mathbf{D}_f)$ for every $h \in VS_{\mathcal{H},\mathbf{D}_f}$ due to the above assumption, including $f \in VS_{\mathcal{H},\mathbf{D}_f}$, by step 1.
3. $\therefore f(\mathbf{x}) = A(\mathbf{x}, \mathbf{D}_f)$.

Comparing Learners with Different Inductive Biases

Rote-Learner. Store examples & classify input instance $\mathbf{x}$ iff it matches that of previously observed example.
No inductive bias.

Candidate-Elimination. Inductive bias: $f \in \mathcal{H}$.

Find-S. Inductive bias: $f \in \mathcal{H}$ and all instances are $-$ve, unless the opposite is entailed by its other knowledge.